MILITARY SPECIFICATION

DIGITAL COMPUTATIONAL SYSTEM FOR REAL-TIME TRAINING SIMULATORS

This specification is approved for use by Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the general characteristics and configurations of digital computational systems used in real-time training simulators, and general guidelines for mathematical models.

2. APPLICABLE DOCUMENTS. There are no applicable documents.

3. REQUIREMENTS

3.0 Computational system. The digital computational system shall consist of a commercially available general purpose digital computer configuration with the options and peripheral equipment specified herein, and associated software (computer programs) to activate and support the simulator. Design of the computational system configuration shall recognize the following life cycle considerations: real-time computer hardware flexibility features, software supportability; ease of computer programming, debug, and reprogramming; quality of support software such as compilers, assemblers, loaders, input/output handlers; maintainability, reliability, and availability. The digital computational system shall accept control inputs, perform a real-time solution of the trainer system mathematical model, and provide outputs necessary to accurately represent the static and dynamic behavior of the real-world systems within the tolerances and performance criteria specified herein. The computational system equipment and computer programs shall have the flexibility and capacity to facilitate programming modifications and additions.

3.1 Computational system equipment. The selected computational system equipment shall be capable of processing the real-time simulation computer program system (CPS).

3.1.1 Computer equipment configuration. An efficient real-time configuration of computer equipment shall be provided. The configuration of the selected computer equipment including central processors, controllers, input/output processors, memory modules, interface multiplexors and signal conversion equipment, and computer peripherals shall be tailored to the real-time simulation information processing requirements and optimized for life cycle expansion capability.

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3.1.1.1 Multiprocessor/multicomputer configuration. If a multiprocessor/multicomputer configuration is proposed to meet the operational requirements of the trainer, the following requirements shall apply to each computer in the system. Multiprocessor/multicomputer configurations shall be designed such that all computers operate in parallel in real-time and are controlled and time synchronized from a single computer program supervisor/executive. This supervisor/executive shall direct the problem flow and establish priority controls. Each computer shall be capable of directly communicating, without involving other central processing units(CPUs), with all peripheral equipment, all instructor station systems for which it computes information or controls information flow, and all interfaces for which it computes or controls information. Each CPU shall be directly loadable independent of other CPUs. All computers in the configuration shall be of the same model and series number. The standard computer vendor equipment shall not be modified. If a multicomputer configuration is proposed to meet the operational requirements of the trainer, each computer shall be capable of transferring or receiving data/information directly to or from other computers. If a multiprocessor computer configuration is proposed, the following requirements shall be satisfied. Each processor shall communicate with a common memory directly accessible to all processors. This common memory shall have capacity sufficient to contain all common information required by more than one processor. This common memory may also serve to buffer data/instructions for swapping between the CPUs and high speed, random access type memory.

3.1.1.2 Computational system performance criteria. The design approach to the computational system configuration must provide efficient real-time capability to process information at the precision, accuracy, stability, and speeds necessary to satisfy the dynamic range of all variables.

3.1.1.3 Computer main frame characteristics. The computer equipment shall provide the memory capacity, word size, speed, and input/output capability compatible with the efficient realization of the total training and support system. The computer equipment shall include the following features (these requirements apply to each computer main frame in the computational system):

a. Power fail safe and automatic restart.

b. Interval timer to measure elapsed CPU time or real-time.

c. Priority interrupts for all peripherals and input/output channels plus two spare interrupts.

d. Hardware bootstrap(s) for prime object loading peripheral magnetic media device(s), including magnetic disc and magnetic tape.

e. Hardware floating-point arithmetic.

3.1.2 Input/output capability. The input/output system of the computer shall have sufficient speed and channel capacity to enable input/output operations to

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proceed in real-time without disrupting or degrading real-time simulation/ operation. To satisfy this requirement, each computer input/output system shall include, but not be limited to, the following capabilities and provisions.

a. Capability to communicate directly with specified digital peripheral equipment.

b. Capability to input or output, while in real-time simulation operation, to one or more units of peripheral equipment, including magnetic disc and tape.

c. Means by which the computer can be externally interrupted.

d. An error checking feature to check input/output transfers.

e. Capability to directly communicate, during real-time operations, with those interfaces for which the computer processes and controls information.

f. Capability to directly communicate with all computers for which data transfer is necessary.

3.1.2.1 Input/output modes. The computer shall have the capability of two basic modes of data transfer as a minimum. The two modes shall be the following:

a. Standard or central processing unit (CPU) controlled input/output

b. Multiplexed input/output.

3.1.2.1.1 Standard or CPU controlled input/output. This input/output mode shall allow each CPU, via specific instructions, to cause data input or output to specific external devices. This mode shall also allow the CPU via other commands to perform control functions with specified external devices.

3.1.2.1.2 Multiplexed input/output. This mode of input/output (I/O) shall allow external devices to transfer blocks of data to or from the computational system memory without CPU intervention for each transfer. This I/O mode shall utilize the existing or standard I/O bus structure of the computer. The blocks of data transferred shall be in continuous locations of memory, and the transfer capability shall be limited only by the size of memory available to the computer. This I/O mode shall be controlled by the CPU only to the extent of data block parameter setup, once per block transfer per channel. This mode of I/O shall allow data transfer on the subsequent instruction cycle following the service request.

3.1.2.2 Interrupts. The computer shall provide the capability to service interrupts in a priority structure. The hardware of each computer shall provide a means of discerning between simultaneous interrupts and provide the computer programs a means of entering service routines for interrupt processing.

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3.1.3 Interface equipment. The interface equipment shall be designed to permit the activation of the trainer stations and sampling of controls with sufficient accuracy and speed to minimize I/O signal error and eliminate discernible discrete stepping of indicators and other appropriate outputs. Standards for form and format of I/O quantities shall be established for the computer side and trainer side of the interface. The interface system shall include analog-to-digital input conversion equipment, digital-to-analog output conversion equipment, discrete inputs, discrete outputs, digital inputs, digital outputs, digital-to-synchro outputs, etc., and necessary control equipment. The interface equipment shall interconnect directly with each computer with which it communicates directly; i.e., the computer in which information originates or terminates.

3.1.3.1 Converters. Digital-to-synchro converters or digital stepping motors shall be used for converting digital information to angular shaft positions. Analog servo equipment shall not be used in the interface equipment.

3.1.3.2 Electronic relays. Electronic relays or switches shall be used for discrete control functions. Mechanical relays shall not be used in the interface system.

3.1.3.3 Converter resolution. The input converters shall be capable of resolutions of one-half the accuracy tolerance specified for the particular control. The resolution of the output converters shall satisfy, as a minimum, the following equation:

$$n = 6 + 3.3 \log l$$

Where:

n = converter word length

l = instrument scale length in inches.

3.1.4 Peripheral and auxiliary equipment. Computer peripheral and auxiliary equipment shall be provided to support the computational system in meeting the requirements of the simulator.

3.1.4.1 Basic peripherals. The following peripheral equipment shall be provided with the computational system and shall be interfaced directly with the computer:

a. Digital remote-control unit (DRU). A digital remote-control unit shall provide the capability to control, monitor, freeze, change register and memory contents, display contents and to halt and restart the simulator operation. The

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DRU shall have the following characteristics:

(1) Capability to address any register and memory location.

(2) Capability to read out the contents of any register and memory location and to change contents of these locations.

(3) Capability to freeze the training situation, alter, and read out the contents of any register/memory location while in this freeze, and continue real-time simulator operation with this unit independent of the computer control panel.

(4) Input capability to assign values to each bit of the basic word.

(5) Capability to operate from any location in the simulator complex, where a signal under computer control can be monitored. The unit shall be portable with a cable and plug which connects into jacks located at appropriate positions around the simulator determined at the mock-up conference.

(6) Readouts availability in octal or hexidecimal format (depending on the computer base) as well as decimal format. Display format shall be selectable by means of a switch control on the DRU control panel. A minimum of two separate indicator registers shall be provided; one to display the address and one to display the contents of the addressed location.

(7) Capability to communicate with all computers in the system by a switch control on the DRU panel.

b. Magnetic disc system. A magnetic disc system shall be the primary system for on-line storage of all simulator computer programs, and the primary system for loading the computer memory with the operational CPS. The number of disc drives and controllers shall be a function of the computational system design, as specified herein, and the total system requirements for on-line real-time information flow and timing between the disc(s) and computer(s).

c. Remote CRT terminal display/keyboard. A remote CRT terminal display/keyboard shall be provided as the primary communication device for computer control. This terminal shall include a cassette type magnetic tape drive system. It shall be possible, during a training mission, to use this terminal to generate a tape which can be later (during non-training time) read by the computer to either update a source file on the disc/input change information to the modification support programs (see 3.2.3.3). This requirement may be satisfied by either a hardware function of the terminal or by a computer program executed in a background mode during a training mission. This terminal shall also include a hard copy device capable of outputting a minimum of 30 characters per second. Display of input data shall be provided on the CRT display system prior to entry into the computational system to allow for correction.

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d. Magnetic tape system. A magnetic tape system shall provide an on-line real-time magnetic storage and access capability. The magnetic tape system shall provide an on-line real-time backup to the disc system. A capability to copy information from magnetic tape onto the disc and to load a new disc shall be provided. The magnetic tape system shall meet the following minimum requirements:

(1) Speed: 45 inches/second

(2) Density: 800 bpi

(3) Nine (9) channels.

The magnetic tape system shall be industry compatible and shall be controlled and computer program configured to provide a backup to the real-time, on-line function of the disc.

e. Alternate peripheral devices. Alternate peripheral devices will be considered if proposed with supporting rationale to establish equivalent performance, capacity, and functional intent of these basic peripherals.

3.1.4.2 Computer program support peripherals. One set of additional peripheral devices shall be specified herein to be included with the computational configuration at the support center (see 3.3).

3.1.5 Real-time clock. A real-time clock(s) shall be provided for timing the operation of the complete computational system. This clock(s) shall be settable under program control, readable under program control, and provide interrupts to the processor(s) of the computational system. The resolution of the clock(s) shall be at least 20 microseconds. This requirement may be satisfied with an interval timer(s).

3.1.6 System spare capacity. Each computer in the computational system configuration shall provide the following spare capacity:

a. Spare computational time shall be verified by running a typical but heavily loaded mission with all systems operational during that mission. The following spare time requirements shall be provided:

(1) The longest computation time utilized during a complete cycle shall be recorded. Spare time shall be provided as defined by:

$$T_1 = \max_{\substack{\text{cycle} \\ j=1}}^{m} \left[\sum_{i=1}^{n} TF_i \right]_j \quad ; \quad T_1 \text{ spare} = \frac{T_{ac} - T_1}{T_{ac}}$$

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Where:

$\max_{j=1}^{m}{}_{cycle}$ = longest time used cycle over m cycles
m = number of cycles sampled
n = number of frames in cycle
TF_i = time used for processing during the ith frame
T_{ac} = available cycle time (1 second of real time).

(2) The maximum time used in any frame shall be recorded. The maximum frame time of these maximums shall be such that spare frame time is provided as follows:

$$T_2 = \max_{j=1}^{m} \left[\max_{i=1}^{n} TF_i \right]_j \quad ; \quad T_2 \text{ spare} = \frac{T_{af} - T_2}{T_{af}}$$

Where:

T_{af} = available frame time; i.e., that period of time defined as inverse of the fastest regular iteration rate for the total real-time computer program.

(a) Step 1: $\max_{i=1}^{n} TF_1$: Compare n frames over a cycle and save the longest frame compute time of the n frames, and

(b) Step 2: $\max_{j=1}^{m}$: Compare and save the maximum of the maximums computed in step 1 over m cycles.

(3) For each computer, the average of the maximum frame time shall be computed with a spare time computed as follows:

$$T_3 = \frac{1}{n} \sum_{i=1}^{n} \cdot \left(\max_{j=1}^{m} \left[\text{sample} \; TF_i \right]_{i=1}^{n} \right)_j / i \; ; \; T_3 \text{ spare} = \frac{T_{af} - T_3}{T_{af}}$$

(a) Step 1: TF_i : Sample and store time for each frame.

(b) Step 2: $\max_{j=1}^{m}$: Compare and store the maximum of each frame over m cycles from step 1.

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(c) Step 3: $\sum_{i=1}^{n}$: Sum the n maximum frame values obtained from step 2.

(d) Step 4: $\frac{1}{n}$: Establish average worst-case frame time.

b. Spare memory shall be provided. The spare memory requirement shall apply to the private memory of each computer in the system and any common memory provided rather than over the total of all memory. Thus by definition the spare percentage of the total memory shall be greater than, or equal to, the total spare requirement. This spare memory shall comprise useable blocks with the minimum contiguous block size of 128 locations.

(1) Spare memory for each computer:

(a) Private spare memory: $\geq M_1$

(b) Common spare memory: $\geq M_2$

(2) Spare memory for the total computational system: $\geq M_3$

c. Spare of each on-line mass storage system shall be provided.

d. Spare capacity of the total input channel, output channel, and each type of interface equipment shall be provided. The input/output system includes the computer I/O and the interface multiplexors, analog-to-digital converters (A/D), digital-to-analog converters (D/A), and all signal distributors, converters, and processors between each computer and the input source and output displays, etc. This requirement is intended to provide direct expansion; i.e., without requiring additional channels and signal processors in the I/O and interface areas. The interface equipment shall include the spare capacity in each cabinet and processing element throughout the total interface system.

3.1.7 System growth capability. The computational system shall be designed to permit expansion of computational capacity without significant design changes to existing hardware and without obsoleting the existing equipment. The following expansion capability shall be provided:

a. Directly addressable memory. 100 percent expansion of the total directly addressable computer system memory. This requirement shall be satisfied by providing the capability to increment memory size by means of the addition of memory modules or, if necessary, to satisfy the processing requirements, the addition of processors with associated memory. A bank switching feature is acceptable to satisfy this requirement.

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b. Mass storage. 100 percent expansion of the on-line mass storage system capacity.

c. Processing capacity. 100 percent increase in the maximum computational processing capacity (instructions/second) by the addition of central processors of the same manufacturer's model and series designation as those selected for the initial system.

d. Input/output. 100 percent increase in the input/output channels and complete interface associated with the computational system.

3.1.8 Physical and environmental characteristics. The computational system shall operate properly, within the total training system requirements, in accordance with the environmental characteristics of the system. The computational system shall operate with the power available at the installation site. Means shall be provided to eliminate transients, pulses, or other electrical noise that would cause computer malfunction.

3.1.9 Computational system availability, maintainability, and reliability. The computational system shall allow the total simulator to meet the availability, maintainability, and reliability requirements specified for the total simulator system.

3.2 Computer program system. A computer program system (CPS) shall be developed to support the total mission of the simulator including training and evaluation of students, maintenance, and support of the total simulator. The simulator CPS shall be organized into the following computer program functional areas:

a. Real-time operational programs:

(1) Supervisor programs

(2) Simulation programs

(3) Input/output programs

(4) Real-time alphanumeric/graphic CRT display programs

(5) Library items.

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b. Simulator support programs:

(1) Verification programs

(2) Preflight check programs

(3) Modification support programs.

c. Computer program system support programs:

(1) Resident operating system

(2) Peripheral operating/handler systems

(3) Loaders

(4) Assemblers

(5) Compilers

(6) Memory dump programs

(7) Mathematical library

(8) Copy routines

(9) Trace routines.

d. Maintenance and test programs:

(1) Computer diagnostic programs

(2) Real-time interface equipment diagnostics.

e. Calibration test programs

3.2.1 Program organization. The total computer program system shall be top-down designed and organized into functional modules such that a function is handled in an independent manner. The modules shall load, assemble, and/or compile independently, such that a modification to a functional module can be accomplished independently. Each module shall be functionally defined and include complete input/output relationships with the computer program and computer equipment. Examples of functional modules include; engines, radio routines, flight, aero, ground dynamics, visual and motion. These functions

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may be further divided into modules such as flight equations of motion, manual flight controls, automatic flight controls, etc. The CPS shall be configuration managed at the module level. It is preferred that structured programming techniques be applied to the design of the CPS to facilitate straightforward comprehensible computer program design. For example, proper programs (i.e., one entry and one exit) structured in nested combinations is a preferred approach.

3.2.2 Real-time operational programs. Programs shall be written to perform the real-time computation and input/output at processing rates sufficient to implement all simulated systems within the specified tolerances and to fully meet the operational training requirements specified herein.

3.2.2.1 Supervisor programs. A supervisor program shall be designed to maintain control over the functional system-oriented and input/output-oriented real-time programs. The supervisor shall control the update of the data base as a result of input/output operations via direct update or by passing information to the other program modules for data base update. The supervisor shall operate at the highest iteration rate and shall control all other real-time modules on a periodic basis within the cycle and frame timing structure of the real-time system.

3.2.2.2 Simulation programs. Programs shall be written to accurately simulate the dynamics of all systems.

3.2.2.2.1 Numerical integration. A method of numerical integration and an integration interval, Delta T, shall be derived or selected for each part of the problem. The method and associated Delta T shall satisfy stability requirements of the problem, as determined by considering at least the critical Eigenvalues or worst-case natural frequencies of the sytem to be solved. The derived or selected integration method and Delta T shall satisfy accuracy requirements of the problem, considering truncation, round off, and propagated errors.

3.2.2.2.2 Interation rates. Interation rates and corresponding input/output rates shall be chosen to assure the following:

a. Model accuracy and stability requirements.

b. Elimination of perceptible time differences between student and simulator response greater than, or less than, that which would occur in the aircraft.

c. Smooth operation of instruments and force systems.

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d. That information processed by the real-world system and perceivable by the student and instructor is not missed by the simulation processing.

e. Compatibility with a visual system.

3.2.2.3 Input/output programs. Programs shall be written for both real-time/non-real-time mapping and control of inputs/outputs with the functional simulation programs. Inputs and outputs shall be processed as single parameters or in block format. Inputs and outputs shall be processed for all subsystems in the simulator system.

3.2.2.4 Real-time alphanumeric/graphic CRT display programs. Operational computer programs shall be developed to implement the information processing, control, storage, and retrieval of all display pages and dynamically changing information.

3.2.2.5 Library items

3.2.2.5.1 Selection of library items. Operational computer programs shall be developed to provide real-time storage and retrieval of on-line computer program items.

3.2.2.5.2 Storage library. Each computer program library shall be stored on a memory storage device compatible with the performance requirements for these library items. As a design goal, all library items shall be stored for real-time access on a single unit of magnetic storage media; i.e., disc pack or magnetic tape. At least one copy of each library shall be provided.

3.2.3 Simulator support programs. Programs shall be developed to support the operational training and use of the simulator. These programs shall include verification programs and preflight programs.

3.2.3.1 Verification programs. Verification programs shall be developed and provided to verify the fidelity of the real-time simulation programs to the approved data. In addition, programs shall be written to verify the operational cycle time and spare time remaining within the cycle and frame periods. Data generated by these programs shall be available to the procuring activity at engineering design reviews. These verification programs shall be designed to run within the deliverable computer configuration.

3.2.3.1.1 Simulation verification programs. The simulation verification programs shall be as specified in 3.3.2.1.4.

3.2.3.1.2 Cycle-time verification programs. A program shall be written to measure the actual frame time durations of each CPU during the real-time execution of the simulation program. This program shall remain a permanent feature of the supervisor program. Output shall be available at the remote

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CRT terminal. The following information shall be provided to demonstrate the three definitions of spare computation time specified herein.

a. The actual cycle execution time of the longest cycle over a test run.

b. The total execution time for all cycles.

c. The average cycle execution time over a test run.

d. The actual frame execution time and frame identity of the longest frame time over a test run.

e. The maximum frame execution time for each frame.

f. The average frame execution times for all frames over a test run.

3.2.3.2 Preflight check programs. A computer program shall be provided to support the operation of the simulator in the normal procedures identified for preflight check. This program shall be provided to easily determine if the simulator is ready for operation. This check program shall operate at normal iteration rates and shall generate a sequence of standard outputs which drive all perceivable outputs and shall recognize all inputs. The sequence of outputs shall be cyclic with provisions to stop the sequence at any of the standard output values. The complete preflight check procedure using this program shall not exceed 15 minutes. This is not intended as a calibration program, but rather as a quick system operational check. Alternate procedures to this requirement will be considered.

3.2.3.3 Mission support programs. Computer programs shall be provided to support the modification of mission data bases which are accessed by the real-time operational programs. These programs shall be written for execution on the delivered computational system configuration. As a minimum, each of the following devices shall be useable to input parameters required by these programs: remote CRT terminal/keyboard, cassette magnetic tape system. As a minimum, programs and procedures shall be provided for the following types of data:

a. Radio/navigation station data

b. Real-time alphanumeric/graphic CRT display.

3.2.3.3.1 Radio/navigation station modification support programs. A computer program shall be developed to facilitate the automatic generation of airfield and radio/navigation station facilities. This program shall accept input data including all parameters identified in the approved design criteria which define airfields and radio stations. The program shall process the input information and output a loadable version of the data formatted as required by the real-time simulation programs using the airfield and radio/navigation information.

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3.2.3.3.2 Real-time alphanumeric/graphic CRT display generation programs. Computer programs shall be developed to generate CRT display pages and the dynamically changing information. These generation programs shall be fully automated and require inputs to define the static information to be formated in the display and the (dynamically changing) information to be displayed in real-time.

3.2.4 Computer program system support programs. The standard support programs specified herein shall be provided. Any proposed changes to these computer program systems shall be fully justified.

3.2.4.1 Resident operating system. A standard resident operating system shall be provided.

3.2.4.2 Peripheral operating/handler systems. Computer program operating/handler systems shall be provided for all peripherals deliverable with the simulator.

3.2.4.3 Loaders. The standard loader programs available from the computer manufacturer shall be provided.

3.2.4.4 Assemblers. The standard assembler programs shall be as specified in 3.3.2.1.1.

3.2.4.5 Compilers. The standard compilers shall be as specified in 3.3.2.1.2.

3.2.4.6 Memory dump programs. These programs shall be capable of dumping the contents of any number of specified memory locations and special registers. The dump program shall be able to dump onto the peripheral devices provided with the computational system including magnetic disc, magnetic tape, and hard copy through the CRT (reference 3.1.4). These programs shall be useable while the simulator is in the training mode as well as in all other simulator modes.

3.2.4.7 Mathematical library. The complete library of standard computer manufacturer mathematical routines shall be provided. In addition, all nonreal-time/real-time mathematical subroutines used in the simulator computer programs shall be included in this library.

3.2.4.8 Copy routines. Utility programs shall be provided to facilitate disc-to-disc, disc-to-tape, and tape-to-disc copying of information in object code as well as source code format.

3.2.4.9 Trace routines. The standard trace routines shall be as specified in 3.3.2.1.3.

3.2.5 Maintenance and test programs. Programs shall be provided to fully test the operation, both of the computational system and the interface equipment.

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When a malfunction occurs in the computer(s) or simulator equipment, these programs shall provide sufficient information to the operator to identify and locate the malfunction. They shall be capable of running with a minimum of operator intervention.

3.2.5.1 Computer diagnostic programs. A complete set of diagnostic programs shall be provided to isolate failures in the digital computer. All diagnostic programs shall be fully automatic to the extent that once the operator has loaded the program and set up initial conditions, the program will automatically generate error indications. Commercially available programs shall be used to satisfy these requirements. Any variation or deviation from these requirements shall be explained and justified. These diagnostics shall check the main frame and main frame options, and all memory units, peripheral units, and arithmetic units. In addition, diagnostics shall be provided to check the input/output handlers and units.

3.2.5.2 Real-time interface equipment diagnostic programs. Programs shall be provided that will enable on-line program control checkout of the simulator interface equipment. They shall be of an automated type requiring a minimum of technician effort and shall provide a hard copy of the test results and identify the malfunctions to the card level as a minimum. These programs shall perform tasks specified herein.

3.2.5.2.1 Discrete input and output test. The real-time interface equipment diagnostic program shall check the proper functioning of all of the discrete input and output channels, including spares, in a closed-loop fashion. All disconnection and reconnection shall be accomplished under program control. The program, upon detecting a malfunction, shall indicate to the operator the failing channel.

3.2.5.2.2 Analog input and output tests. The analog input and output tests shall exercise all of the analog and synchro devices through their full range of operation. This shall be accomplished in a closed-loop fashion using a known accurate and calibrated digital-to-analog converter as a reference. All disconnections and reconnections shall be accomplished under program control. The test shall be designed so that an accuracy limit can be specified by the operator. It shall test all converters, multiplexors, and demultiplexors, including spares, and all other signal processors in the interface system. All channels not functioning within the specified limits shall be indicated as such. A dynamic test shall enable the operator to vary the period and amplitude of the test signal to the specified channel via an on-line input device.

3.2.5.2.3 Program control test of real-time clock. This program shall automatically check the clock for accuracy and proper functioning and output results via an on-line output device, such as the CRT terminal.

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3.2.5.2.4 Simulator equipment test. Programs shall be provided to check the proper functioning of the remaining simulator equipment and shall output the results via an on-line output device. This program shall test, as a minimum, the following equipment:

a. Input/output controllers, external to the computer.

b. Special purpose peripheral equipment included, i.e., the DRU.

c. Special purpose display processing equipment for the cockpit.

d. Aural/communication processing equipment that is accessible by the software for testing in an automatic feedback manner.

3.2.6 Calibration test programs. Calibration test programs shall be provided to check the accuracy and flow of signals, both statically and dynamically through the full range of variables, between the computer and all signal sources and terminal points. This program shall also be used to calibrate the interface equipment, displays and controls, and to determine whether the inputs provided by the simulation program are being transmitted properly to the ultimate destination.

3.2.7 Deliverable computer program format and media. The CPS shall be delivered in object format on magnetic disc and magnetic tape. Additional CPS media, to be included with the computational system at the support center, shall be specified under 3.3.

3.3 Support requirements. One simulator system as defined herein shall be developed and configured with additional computer equipment and computer program features.

3.3.1 Support computer equipment. One set of the following peripheral equipment and computer main frame options, including controllers and devices, shall be provided:

a. Card reader: Minimum read speed shall be 300 cards per minute.

b. Card punch: Minimum punch speed shall be 100 cards per minute.

c. Line printer: Minimum print speed shall be 600 lines per minute.

d. Key punch: This device shall include a punched card interpreter capability.

e. Additional magnetic disc unit: This unit shall provide the capability for disc-to-disc copy, and rapid and efficient computer program edit/reconfiguration.

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The additional magnetic disc unit shall be identical to the primary magnetic disc specified under basic peripherals; (see 3.1.4.1).

f. Program restrict/instruction trap.

g. Address trap.

h. Hardware bootstrap for the card reader.

The card reader/punch, line printer; and additional disc unit shall be interfaced directly with the computational system. If a multiprocessor/multicomputer configuration is employed, each peripheral equipment controller and device shall interface directly with each computer. A combination card reader/card punch shall be acceptable to satisfy 3.3.1a and 3.3.1b above. The additional main frame options shall be provided for each main frame in the computational system.

3.3.2 Support programs. Support programs shall be provided as specified herein to facilitate computer program generation; redesign, update, modification, error corrections; and support functions at the support center. The support programs to be provided shall include and be organized as follows:

a. CPS support programs

b. Simulation system update and modification programs.

3.3.2.1 CPS support programs. Assemblers, compilers, and verification programs shall be provided to facilitate CPS support.

3.3.2.1.1 Assemblers. The standard assembler programs available from the computer manufacturer shall be provided.

3.3.2.1.2 Compilers. The standard FORTRAN package available from the computer manufacturer shall be provided. In addition, any other standard specifically developed compilers used to develop the simulator computer programs shall be provided.

3.3.2.1.3 Trace routines. The standard trace routines (used for computer program debug) available from the computer manufacturer shall be provided.

3.3.2.1.4 Simulation verification programs. A generalized program(s) shall be written to perform the source-code verification task. As a minimum this program(s) shall be capable of reading and operating on the following inputs and parameters:

a. Module name, variable names; and constant names and their applicable values.

b. Desired output device and format (capability to output on any peripheral; including support peripherals as defined herein):

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c. Names of variables whose values are to be read as input data and names of variables whose values are to be output.

d. Iteration rate for the modules to be tested.

This program(s) shall be capable of linking with the module to be tested, setting-up appropriate conditions for module execution in conjunction with input parameters, reading input data, exercising the module, and outputting the results of the test for comparison with the approved data.

3.3.2.1.5 Modification support programs. Computer programs used to design and implement operational data bases shall be provided to facilitate modification of these data bases. Computer programs used to perform data reduction to tabularized and/or stored data shall be provided. These programs shall be written for execution on the computational system configuration designed for the simulator support site.

3.3.2.2 Simulation system update and modification programs. A program shall be provided to efficiently edit the source code in order to maintain CPS configuration control and to support CPS modification and update. Source EAM cards/disc source images shall be useable as the baseline source code media. This procedure shall support selective module editing, compiling/assembling, linking, and testing. This procedure shall be useable by simulator maintenance technicians. The input device for change information shall be the remote CRT terminal keyboard/cassette magnetic tape system.

3.3.2.3 Deliverable computer program format and media. The CPS shall be delivered in object format on magnetic disc and magnetic tape. The CPS shall be provided in source code format on EAM cards.

3.4 Applicable terms. The definitions of terms as specified herein shall apply to this specification.

3.4.1 General purpose digital computer: A general purpose digital computer is a stored program, automatic-sequence, commercially available digital computer prossessing a memory randomly accessible at the basic machine cycle time (for both instructions and data), general input/output capability, and the ability to perform instructions in an arbitrary sequence which can be self-modified.

3.4.2 Processor word length: Processor word length in the number of bits used in the general purpose registers of the central processing unit to perform single precision fixed-point arithmetic and logical computations. This length does not include bits used for parity check, error correcting, or other non-computing bits.

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3.4.3 Floating-point instructions: Floating-point instructions shall be hardware fixed-wire, floating-point instructions as opposed to programmed floating-point operations.

3.4.4 Simulator systems: Simulator systems consist of the equipment, operational or simulated, which provide the displays, motions, and controls at which the students, instructors, and operators act upon or communicate with the training simulator.

3.4.5 Interface equipment: Interface equipment is the equipment that controls, transmits, encodes, decodes, converts, or buffers analog, digital, or discrete information passing between the computer and the simulator systems; i.e., digital-to-analog, analog-to-digital, digital-to-discrete, and discrete-to-digital converters.

3.4.6 Computational system: The digital computational system shall consist of all computer equipment in the information processing configuration including computer mainframes, memory, input/output processors, peripheral equipment, and interface equipment. In addition, all deliverable computer programs shall be included in the computational system.

3.4.7 Computer program system: The computer programs required to operate, test, maintain, and support the training system.

3.4.8 Computer program documentation: The descriptive material required to define the information content and organization of the computer programs. This includes narrative descriptions of the system to be simulated, implemented programs, and math model flow charts, etc. (Reference 6.2).

3.4.9 Computer program media. The paper and magnetic media which contains the computer program instructions and data. Computer program media must be compatible with the input peripheral devices.

3.4.10 Computer program support: The logistic procedures and resources necessary to accomplish simulator life cycle support through maintenance and modification of the simulator CPS. This includes a configuration management system.

3.4.11 Computer program verification: Determination of the accuracy of the developed computer program relative to the specified requirements and tolerances and the approved design criteria.

3.4.12 On-line: Interconnected with the computational system and under computer control.

3.4.13 Off-line: Not connected with the computational system.

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3.4.14 Simulator operational time: That period of simulator operation including preflight, training mission, and post mission printout or dump.

3.4.15 Simulator CPS: All computer programs associated with a particular simulator including computer programs used during development, programs incorporated into the simulator to support the operation requirements, and programs developed and used to support and maintain the training system.

3.4.16 Structured programming techniques: Structured programming techniques used to facilitate straight forward design and implementation of computer programs. Included in these techniques are top-down design, modular organization and the implementation of proper programs. A proper program is defined as a program which contains only one entrance and one exit.

3.4.17 Multiprocessor: A multiprocessor shall be defined as a computer equipment configuration prossessing more than one central processing unit which utilizes multiported common memory. This memory is directly accessible to all processors and is used for processor communication or as a buffer for I/O data transfer.

3.4.18 Multicomputer: A multicomputer shall be defined as a computer equipment configuration which utilizes standard high speed I/O data channels for processor communication.

3.4.19 Nonreal-time computer programs: Nonreal-time computer programs are those programs that are delivered as part of the Computer Program System and are used to support the simulator while the training mission is not in progress.

3.4.20 Background computer programs: Background computer programs are those programs that are executed in the background mode, of the operating system being used, while the training mission is in progress. These programs are not time critical and as such their processing can be interrupted and carried over from one cycle (one second of real-time) to the next. For the purpose of this standard, background computer programs shall be defined as a subset of the real-time computer programs.

3.4.21 Real-time computer programs: The computer programs which implement the simulated systems, control the simulation processing (executives, input/output, aircraft systems, instructor functions, etc) and all other programs which are executed while the training mission is in progress.

4. QUALITY ASSURANCE PROVISIONS.

4.1 General. Quality assurance activities shall be conducted by the contractor, by the procuring activity, and by designated representatives of the procuring activity.

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4.2 Test criteria. Tests shall be conducted in accordance with an approved Acceptance Test Procedures (ATP) document which will be prepared by the simulator contractor. The ATP shall include tests based on design criteria, and all requirements of this standard.

4.3 Specific tests. The computational system shall be comprehensively tested in accordance with table I requirements (paragraph 4.4). The tests to be conducted shall be developed by the simulator contractor and presented in the ATP, and shall include, but not be limited to, specific tests as identified in the following subparagraphs.

4.3.1 Computational system. The computational system requirements identified in paragraph 3 and associated subparagraphs shall be verified according to the methods identified in the matrix verification table specified herein. Step-by-step procedures shall be prepared in the acceptance test procedures for all computation requirements to be verified by demonstration and test. Verification requirements for the computational system shall include following specific areas of coverage. The procuring activity shall have the right to conduct any additional procedures deemed necessary to verify the computational system requirements.

4.3.2 Computational system performance criteria. Verification procedures shall be written to verify the capability to process information at the precision, accuracy, stability, and speeds to satisfy the dynamic range of all variables.

4.3.3 Multiprocessor/multicomputer system. If a multiprocessor/multicomputer configuration is used, tests and procedures shall be provided to verify the requirements stated in 3.1.1.1 of the specification.

4.3.4 Power fail-safe and automatic restart. A test shall be conducted to verify the operation of the power fail-safe and automatic restart. The verification shall assure continuity of program execution and simulator operation in the event of a power shutdown, outage, or variation not within the computer manufacturer's tolerable range of input power.

4.3.5 Input/output system. Tests shall be conducted to accurately verify the input/output system. These tests shall include, but not be limited to, tests of standard or CPU controlled input/output, multiplexed input/output, interrupt operations, and input/output interface.

4.3.6 Basic and support peripheral. Tests shall be prepared to accurately verify the operating requirements of the following peripheral devices.

a. Digital remote-control unit

b. Magnetic disc system

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c. Remote CRT computer terminal display/keyboard/cassette magnetic tape unit/hard copy unit

d. Magnetic tape system

e. Support peripherals.

4.3.7 System spare capacity. Tests shall be conducted to accurately verify the computational spare capacity specified herein. Spare computational time shall be verified by a typical, but heavily loaded mission with all systems operational during the mission. In addition, tests shall be conducted to verify the percent spare of direct address memory, on-line mass storage memory, total input/output channels, and interface equipment.

4.3.8 System growth capability. The system growth capability specified herein shall be verified to procuring activity personnel by computer manufacturer's date, cross-referenced to the actual equipment.

4.3.9 Computer program organization verification. The contractor shall develop and perform procedures to verify that the total computer program system is modularly organized such that each module can be independently assembled/compiled and loaded. Verification shall also be provided to verify that modification to each module can be accomplished independently.

4.3.10 Computer program system verification. The contractor shall develop test procedures for the computer program system which is specified under 3.2. The procuring activity shall have the right to perform any additional tests deemed necessary to assure that the computer program system (CPS) conform to the prescribed requirements.

4.3.10.1 Real-time operational program verification. Tests shall be conducted to verify that the real-time computations and input/output processing rate is sufficient to implement all simulated systems within the specified tolerances and still satisfy the system spare requirements.

4.3.10.1.1 Real-time alphanumeric/graphic CRT display generation programs. Tests shall be conducted to verify the operation of the programs developed to generate and implement the information processing, control, storage, and retrieval of all display pages and dynamically changing information specified herein.

4.3.10.1.2 Library items. Tests shall be conducted to verify the operational programs developed to provide real-time storage and retrieval of on-line computer program items.

4.3.10.2 Simulator support programs verification. Verification of the simulator support programs (as identified in 3.2.3), including verification programs and preflight check programs, shall be conducted.

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4.3.10.3 Computer program system support programs verification. Tests shall be conducted to verify the computer program system support programs which are identified in 3.2.4.

4.3.10.3.1 Memory dump programs. Tests shall be conducted to verify the operation of the memory dump programs. These tests shall test the capability to dump specified locations of memory onto magnetic tape, magnetic disc, and hard copy through the CRT.

4.3.10.3.2 Computer diagnostic programs. The complete set of diagnostic programs shall be exercised to verify the correct operation of the computer diagnostic programs in isolating failure in the digital computer system for all possible internal operations and computer dependent operations. These diagnostics shall be exercised following the prescribed procedures identified in the diagnostic operations manuals to insure agreement between the program operation and documentation.

4.3.10.3.3 Real-time interface equipment diagnostics. Tests shall be conducted to verify the operation of the real-time interface equipment diagnostics. These diagnostic tests shall include verification of all discrete input and output channels, analog and synchro devices, real-time clocks, and all remaining simulator equipment. These tests shall produce hard copy output via on-line output devices.

4.3.10.3.4 Calibration test programs. Tests shall be conducted to verify the operation of the calibration test programs.

4.3.11 Simulation system update/modification. The contractor shall develop tests to verify the capabilities specified in 3.3.2.2 for system flexibility. Tests shall be developed to verify the capability for selective editing of the source program modules. These tests shall demonstrate the ability to create input for the editor on the computer terminal's cassette magnetic tape unit during a real-time simulation exercise and then the use of this tape after the simulation exercise to rapidly edit and test a program module.

4.4 Test matrix. The test matrix (table I) establishes the method of verification/qualification to be applied by the procuring activity or its designated representatives. Each column heading of table I shall be defined as follows:

a. Requirement: This column lists the applicable paragraphs of section 3 herein to be verified or other requirements of sections of reference documents as identified in table I.

b. Not applicable (N/A): This column will show an X if the particular paragraph cannot or will not be verified.

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TABLE I (Page 1 of 2 Pages). Test Matrix

REQUIREMENT	N/A	INSP	ANAL	DEMO	TEST	REQUIREMENT	N/A	INSP	ANAL	DEMO	TEST
3.				x	x	3.2.2				x	x
3.1			x	x		3.2.2.1				x	
3.1.1		x		x	x	3.2.2.2			x		x
3.1.1.1		x		x	x	3.2.2.2.1			x		
3.1.1.2					x	3.2.2.2.2			x		x
3.1.1.3			x		x	3.2.2.3				x	
3.1.2					x	3.2.2.4				x	x
3.1.2.1	x					3.2.2.5	x				
3.1.2.1.1					x	3.2.2.5.1				x	x
3.1.2.1.2					x	3.2.2.5.2					x
3.1.2.2					x	3.2.3	x				
3.1.3				x	x	3.2.3.1	x				
3.1.3.1		x	x			3.2.3.1.1	x				
3.1.3.2		x	x			3.2.3.1.2					x
3.1.3.3			x			3.2.3.2					x
3.1.4	x					3.2.3.3			x	x	x
3.1.4.1		x		x	x	3.2.3.3.1					x
3.1.4.2	x					3.2.3.3.2					x
3.1.5					x	3.2.4	x				
3.1.6			x	x	x	3.2.4.1				x	
3.1.7			x			3.2.4.2				x	
3.1.8		x			x	3.2.4.3				x	
3.1.9				x		3.2.4.4	x				
3.2			x			3.2.4.5	x				
3.2.1					x	3.2.4.6				x	

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TABLE I (Page 2 of 2 Pages). Test Matrix

REQUIREMENT	N/A	INSP	ANAL	DEMO	TEST	REQUIREMENT	N/A	INSP	ANAL	DEMO	TEST
3.2.4.7				x		3.4.2	x				
3.2.4.8				x		3.4.3	x				
3.2.4.9	x					3.4.4	x				
3.2.5					x	3.4.5	x				
3.2.5.1					x	3.4.6	x				
3.2.5.2					x	3.4.7	x				
3.2.5.2.1					x	3.4.8	x				
3.2.5.2.2					x	3.4.9	x				
3.2.5.2.3					x	3.4.10	x				
3.2.5.2.4					x	3.4.11	x				
3.2.6					x	3.4.12	x				
3.2.7		x				3.4.13	x				
3.3	x					3.4.14	x				
3.3.1		x		x		3.4.15	x				
3.3.2	x					3.4.16	x				
3.3.2.1	x					3.4.17	x				
3.3.2.1.1				x		3.4.18	x				
3.3.2.1.2				x		3.4.19	x				
3.3.2.1.3					x	3.4.20	x				
3.3.2.1.4					x	3.4.21	x				
3.3.2.1.5			x	x	x						
3.3.2.2			x	x	x						
3.3.2.3		x									
3.4	x										
3.4.1	x										

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c. Inspection (INSP): A visual inspection by the responsible in-plant representative or the procuring activity test engineering personnel for verification of having satisfied the requirement.

d. Analysis (ANAL): Analysis of documented data as verification of having satisfied the requirement. The actual analysis may be performed by contractor personnel, but approval/disapproval authority lies with the procuring activity.

e. Demonstration (DEMO): Demonstration as verification of having satisfied the requirements shall be by the contractor for either the responsible in-plant representative, procuring activity test engineering personnel, or both.

f. Test: Test as verification of having satisfied the requirement shall be hands-on testing conducted by the responsible in-plant representative, procuring activity test engineering personnel, or both. Any demonstration or test indicated in table I may be supplemented with analytical verification, at the option of the procuring activity. The ATP document shall include verification procedures for all table I requirements.

5. PACKAGING

5.1 Packaging, as applicable, shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The requirements of this specification are intended to establish a baseline in developing standardization for USAF flight simulator computational systems.

6.2 Data requirements. Data requirements in support of this specification will be in accordance with DI-H-3277 and will be as reflected in a Contractor Data Requirements List (DD Form 1423).

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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